



4. Evidence Preservation

Objective

Practice evidence collection, hashing, and chain-of-custody documentation.

Tools Used

Velociraptor

FTK Imager

Task 1 – Volatile Data Collection

Environment Details

Component	Details
Operating System	Windows 11
Tools Used	Velociraptor, FTK Imager
Analyst Role	SOC Analyst
Collection Type	Volatile Data & Memory Acquisition
Hashing Algorithm	SHA-256

Purpose

Volatile data contains information stored in memory that may be lost after a system reboot or shutdown. Collecting this data helps investigators identify:

- Active network connections
- Suspicious remote sessions
- Malware communication

Procedure Performed

Step 1: Launch Velociraptor

The Velociraptor client was opened on the Windows VM to initiate forensic artifact collection.



Procedure Performed

Step 1: Launch Velociraptor

The Velociraptor client was opened on the Windows VM to initiate forensic artifact collection.

The screenshot shows the Velociraptor web interface. At the top, there's a search bar with 'recent:admin' and a user dropdown set to 'admin <root>'. Below the search bar is a toolbar with icons for adding, deleting, and editing notebooks, along with pagination controls showing '0-1/1' and '10'. A table lists the available notebooks:

NotebookId	Name	Description	Creation Time	Modified Time	Creator	Collaborators
N.D8231R7QTTI0K	Evidence Preservati	Collecting volatile data and	2026-05-13T08:06:04Z	2026-05-13T08:06:04Z	admin	

Below the table, a message says 'Select a notebook from the list above.'

Step 2: Execute Netstat Query

The following query was executed to collect active network connection details:

```
SELECT * FROM netstat
```

The screenshot shows the Velociraptor web interface with a new notebook 'N.D8236ERJVCQ0' selected. The notebook name is 'New Notebook', created on '2026-05-13T08:15:55Z'. Below the notebook list, a query editor shows the executed query 'SELECT * FROM netstat'. The results are displayed in a table with columns: Family, Type, Laddr, Raddr, Status, Pid, FamilyString, and Timestamp.

Family	Type	Laddr	Raddr	Status	Pid	FamilyString	Timestamp
2	1	{ "IP": "0.0.0.0", "Port": 135 }	{ "IP": "0.0.0.0", "Port": 0 }	LISTEN	1468	IPv4	2026-05-13T07:14:04Z



Step 3: Data Collection

Velociraptor gathered:

- Local IP addresses
- Remote IP addresses
- Connection states
- Protocols (TCP/UDP)
- Listening ports
- Associated processes

Step 4: Export Results

The output was exported and saved in CSV format for later forensic analysis.

Fd	Family	Type	Laddr	Raddr	Status	Pid	FamilyStr	Timestamp	TypeString
0	2	1	{	{	LISTEN	1468	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	4	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	4	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	5144	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10648	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10744	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	12908	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	8684	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	10472	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	3328	IPv4	2026-05-1	TCP
0	2	1	{	{	LISTEN	6156	IPv4	2026-05-1	TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1	TCP

3. Memory Acquisition

Purpose

Memory acquisition captures the contents of RAM for forensic analysis. This helps identify:

- Running malware
- Injected code
- Active user sessions
- Encryption keys
- Network artifacts
- Malicious processes

Procedure



Step 1: Open Task Manager

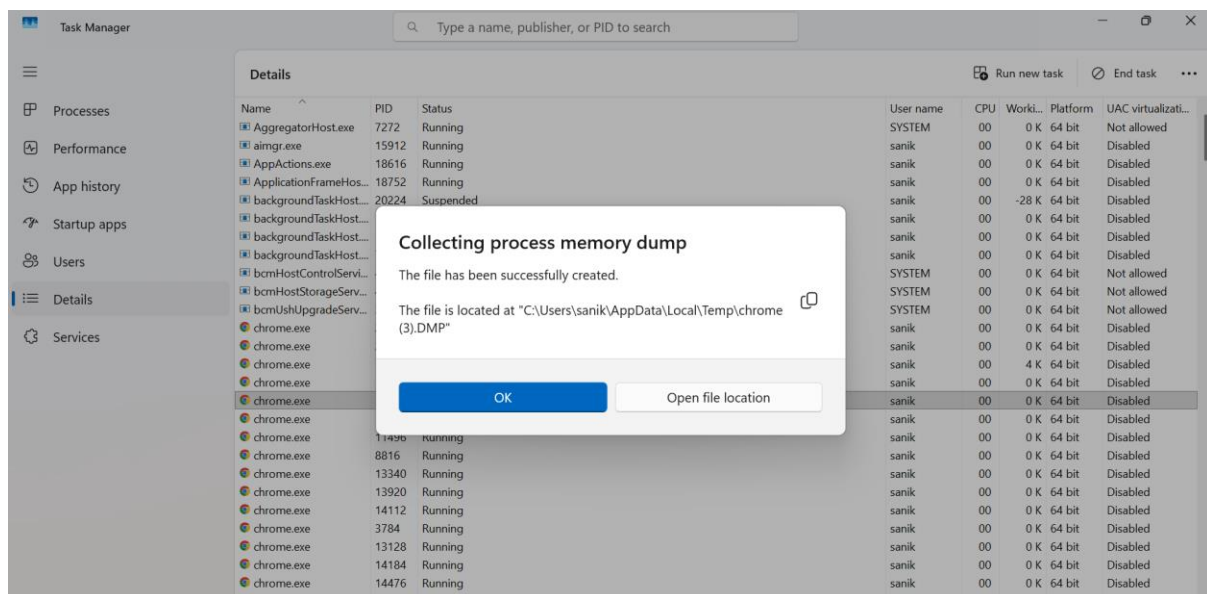
- Press Ctrl + Shift + Esc
- Open the Details tab

Step 2: Select Process

- Identify the target process (example: lsass.exe, explorer.exe, or suspicious process)

Step 3: Create Dump File

- Right-click the process
- Select Create dump file



Step 4: Dump File Location

Windows saves the dump file in:

C:\Users\sanik\AppData\Local\Temp\chrome.DMP

Step 5: Hash the Dump File

Open Command Prompt in the dump location and run:



```
Windows PowerShell (x86) X + v
PS C:\Users\sanik> Get-FileHash "C:\Users\sanik\AppData\Local\Temp\chrome.DMP" -Algorithm SHA256

Algorithm      Hash                                          Path
-----
SHA256         22FD04A44F613349B376B9329B2E637B2DC5B4DDA14346CF27B54153DFC04D13  C:\Users\sanik\AppData\Local\Temp\chrome.DMP

PS C:\Users\sanik> |
```

Chain of Custody Documentation

Purpose

Chain of custody ensures that evidence remains authentic, untampered, and legally admissible.

Evidence Preservation and Chain of Custody

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Evidence Preservation and Chain of Custody

Item	Description	Collected By	Date	Hash Value
Memory Dump	Chrome Process Dump	Soc Analyst	2026-05-13	22FD04A44F613349B376B9329B2E637B2DC5B4DDA14346CF27B54153DFC04D13
Network Connections	Netstat CSV Export	Soc Analyst	2026-05-13	CSV Export Generated